Nissai High Altitude Research Site, Western Pamir, Tajikistan

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Background

Within SNF project “Precipitation in extreme environments” the SLF has partnered with UNIFR (Hoelzle/ Saks) under the SDC funded project CROMO-ADAPT (Cryosphere Monitoring and Observations for Adaptation in Central Asia) and collaborates with the Centre for Research of Glaciers (CRG) in Dushanbe to conduct scientific research in the high mountain areas of Tajikistan. Fieldwork was undertaken in the Nissai range, Sangvor district, which lies in the upper Khingob watershed. The aim was to install a high mountain monitoring station with a focus on high altitude precipitation. We also installed a network of surface temperature thermistors in order to investigate permafrost distribution in the region. Sangvor district has been proposed as it contains the Khingob river which is the second main tributary of the Vakhsh. It is therefore hydrologically very significant and currently unmonitored. Together with the WSL HIMAL station in Jirgatol we will have a good basis for basin level investigations. The station itself has been jointly financed between the CRYOS lab at EPFL and SLF with financing contributions from the SNF. The station was deployed for one season in Dischma, Davos (2020-2021) for testing as well as contributing to several ongoing research projects.

Expedition aims

1. Establish a high-altitude automatic weather station with snow parameters and micrometeorological measurements.
2. Establish surface temperature network as basis for permafrost monitoring.
3. Start collaboration with CRG scientists (however field participation was unfortunately not possible this time due to overlapping expeditions).

Location

The expedition took place from 18 - 26 September 2021 in the Nissai range which lies between the upper Khingob River and Ab-i- Mazor. Scientists from the WSL Institute for Snow and Avalanche Research SLF collaborated with the Centre for Research of glaciers led by Prof. Kayumov. The expedition was conducted within the framework of the pre-phase CROMO-ADAPT project jointly led by UNIFR (Hoelzle / Saks) and SLF (Fiddes).

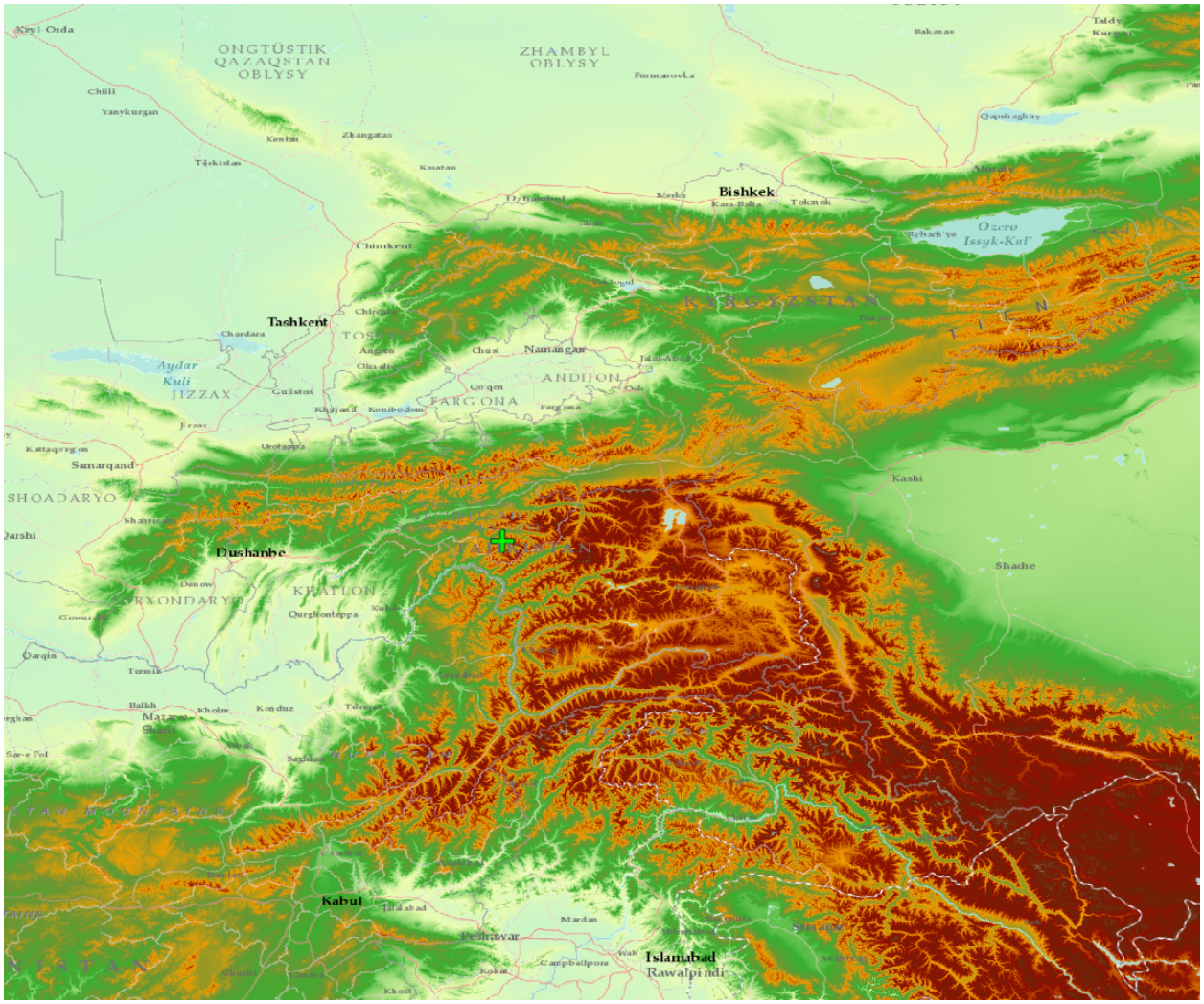


Figure 1: Nissai station location in the Western Pamir, Tajikistan (green cross).

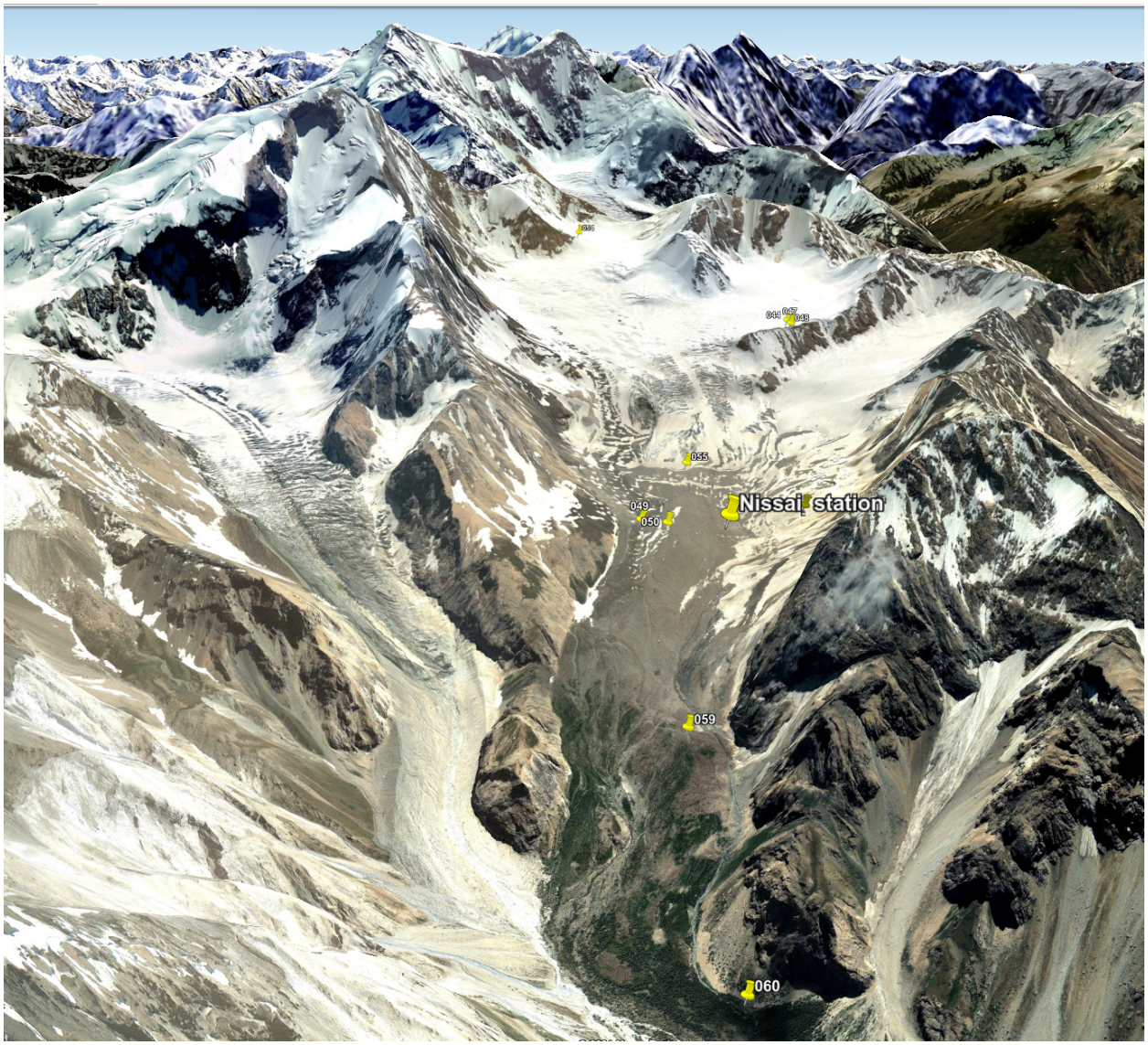


Figure 2: Google earth view of the Nissai station catchment with station location and surface temperature logger network.

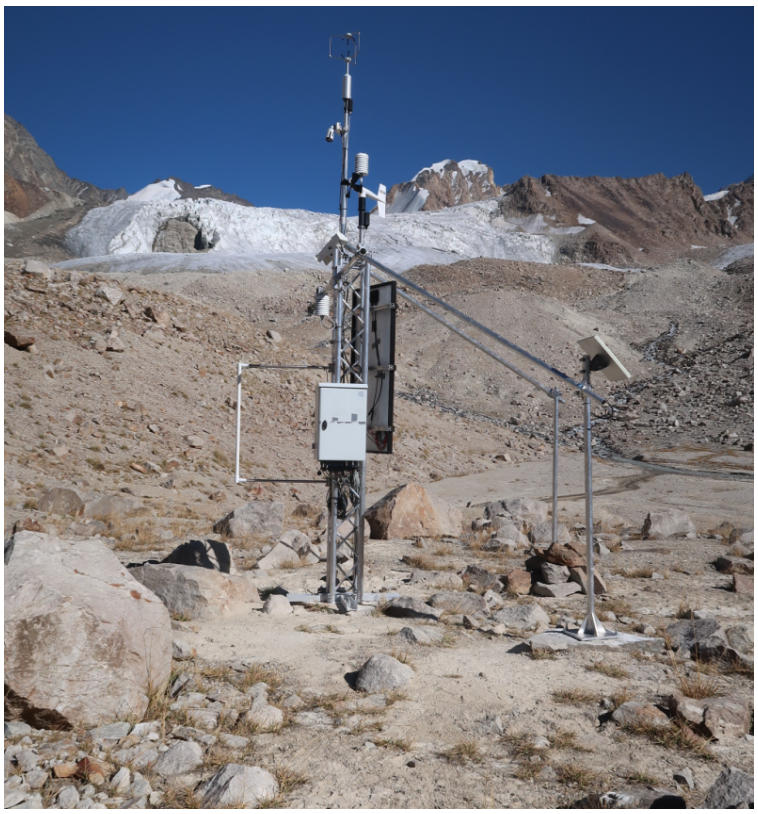


Figure 3: Nissai AWS location.

Automatic weather station

38.74287, 71.35392

The automatic weather station is situated in a moderately exposed location on a small rise above the proglacial plain at 3909m asl at 38.74287N, 71.35392E. It measures standard meteorological variables as well as specific instruments such as the sonic anemometer that enable us to investigate boundary layer meteorology. Additionally we measure snow height and lateral fluxes (snow drift) with a FlowCapt. With this station we will be able to gain detailed information about the climate at this high altitude site and perform detailed numerical simulations using these meteorological observations. It will also provide a basis to evaluate large scale basin investigations. Variables measured at the station are:

- Sonic anemometer (3D wind)
- Multilevel air temperature /Rh naturally ventilated
- Wind propeller (speed and direction)
- SW/LW up/down

- IR Surface temperature
- Laser snow depth
- Snow drift flux
- 10/50cm ground temperature
- Liquid precipitation



Figure 4: constructing the Nissai station.

Table 1: Detailed list of installed instruments on the AWS

Sensor	Variable	Height (cm)
Apogee/Campbell SN-500-SS	Radiation (SW/LW)	273
Apogee/Campbell SI-111-L-10 (TSS)	Surface temperature	273
Campbell 107 soil probes	soil temperature	soil1: -10cm, soil2: -50cm
Young 3D Sonic Anemometer 81000RE	Wind direction and speed	443
Campbell CS215, with RAD10	Air temperature/ Humidity	243
Campbell HygroVUE10, with RAD10	Air temperature/ Humidity	373

Campbell ClimaVUE50	mini station (TA,RH, wind, precip)	273
Campbell CR1000X logger plus case	data logger	
FlowCapt FC4 mounting kit	drifting snow	
Lufft SHM31	snow depth	273
LeadCrystal 6-CNFJ-70 battery	battery	
Camera, Milesight MS-C5364-PB-J MiniBullet 2.8mm/H113° 5MP	camera	
Hughes 9502 Modem, antenna and cable	Station status monitoring	180
Young/Campbell 05103-5 (propeller)	wind direction and speed	313
FlowCapt FC4	wind drift	110-210
main mast	mast	
2m aluminium pole	mast	
3m aluminium poles	mast	
2m aluminium pole plus plate	mast	
Solarpanel, AxSun AX M-36 185	power	

Satellite
EMEA
239.2 bearing
24.6 Azimuth

Calibrate laser snow depth
191 deg bearing
45 deg angle

Solar panel mounted at 180 degrees (South)

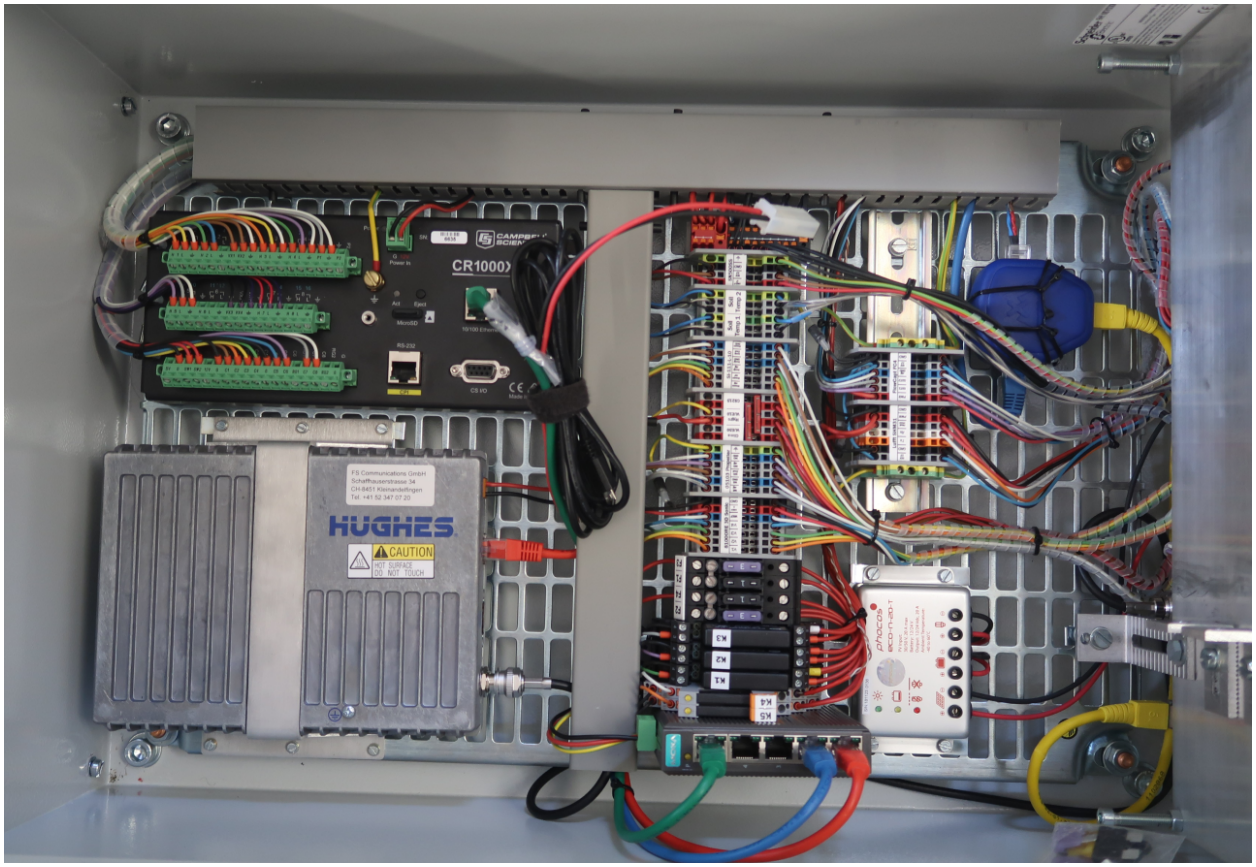


Figure 5: Inside the logger box

Permafrost network

Permafrost is ground material that remains permanently frozen, it can be ice rich but doesn't necessarily have to be. To directly measure permafrost requires a borehole equipped with thermistor chains. However a suitable technique for monitoring permafrost (in addition to boreholes) that is used in the Swiss national Permafrost monitoring network (PERMOS) are small ground surface temperature loggers which are embedded in rock fissures or ground material to avoid direct solar radiation. A mean annual ground surface temperature (MAGST) of less than 0°C can be used as a proxy for Permafrost existence, or at least a surface climate forcing that would allow permafrost to develop.

The loggers we use are (1) SLF designed BLE loggers that we communicate with via a smartphone application and bluetooth connection and (2) Geoprecision rock-loggers. Both loggers are capable of making hourly measurements for 7-10 years on a single battery so suitable for long term monitoring networks.

An elevation transect of loggers was installed from the start of the Nissai valley (~2400m asl) up to the upper glacier basin (~4700m asl) so that we have over 2000m elevation range. These loggers will store hourly values of the surface temperature in various mediums (debris/ bedrock) as a proxy for permafrost occurrence). Additionally, these loggers will allow us to evaluate new permafrost model based maps for the region. Finally we will be able to investigate the effect of exposure to solar radiation on the ground thermal regime on north/south slopes with this data.



Figure 6: BLE temperature logger smartphone app and GPS (left), Geoprecision logger (right).

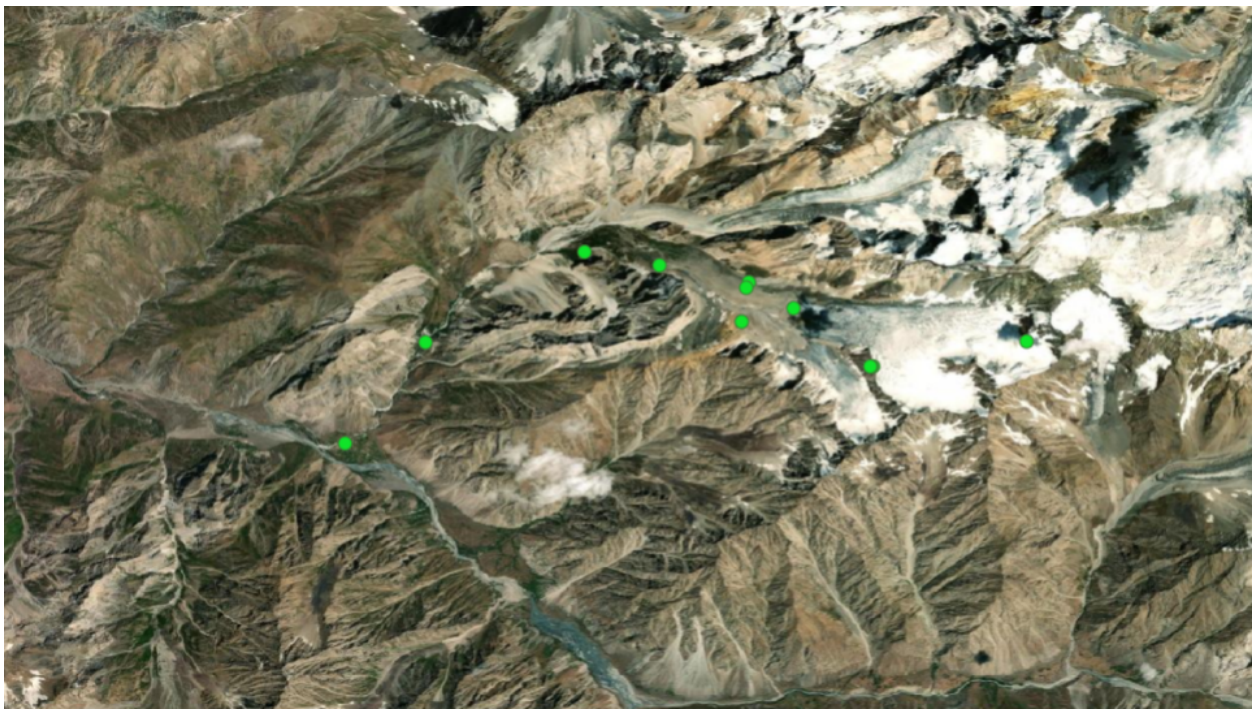


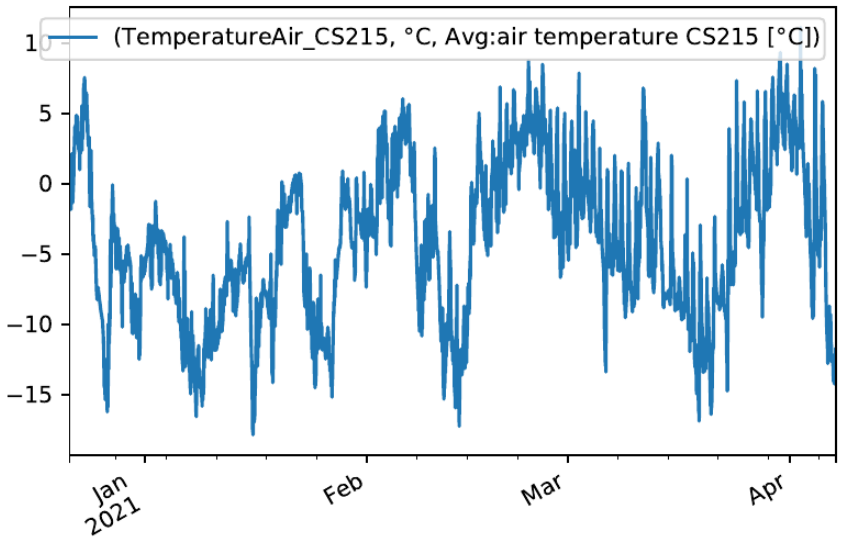
Figure 7: The locations of the installed permafrost network sensors given in green.



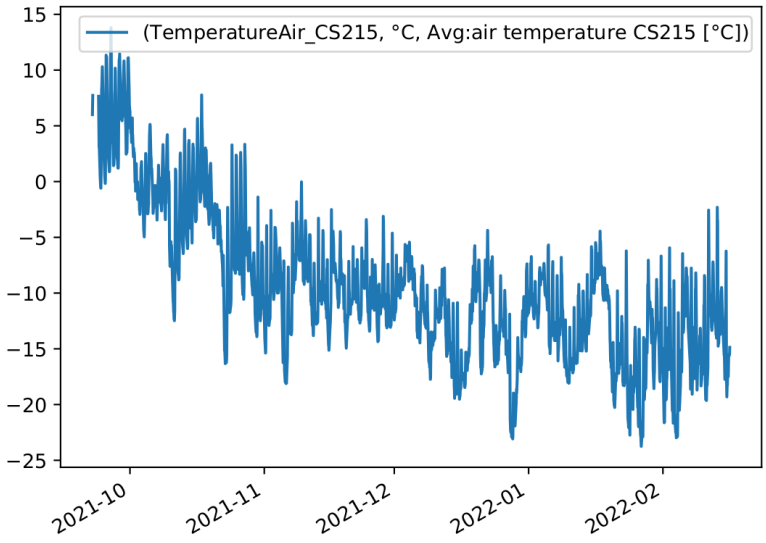
Figure 8: Permafrost sensor site on upper glacier.

Appendix 1: Example of station parameters from testing in Switzerland 2021 and deployment Tajikistan 2021-2022

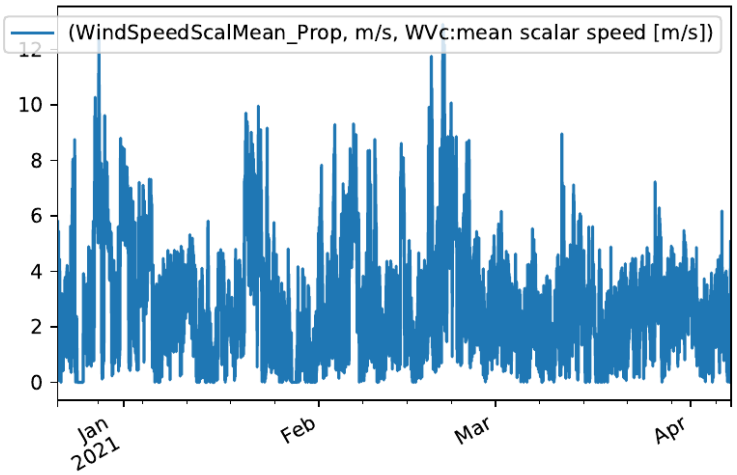
Air temperature DAV



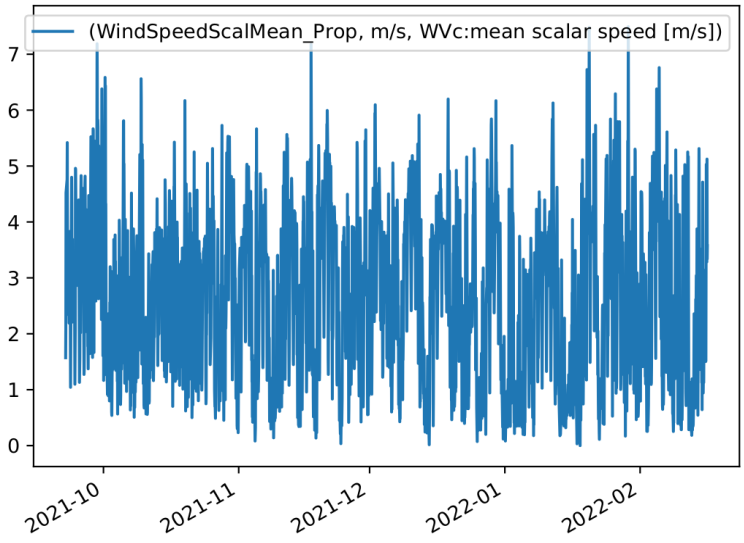
Air temperature TAJ



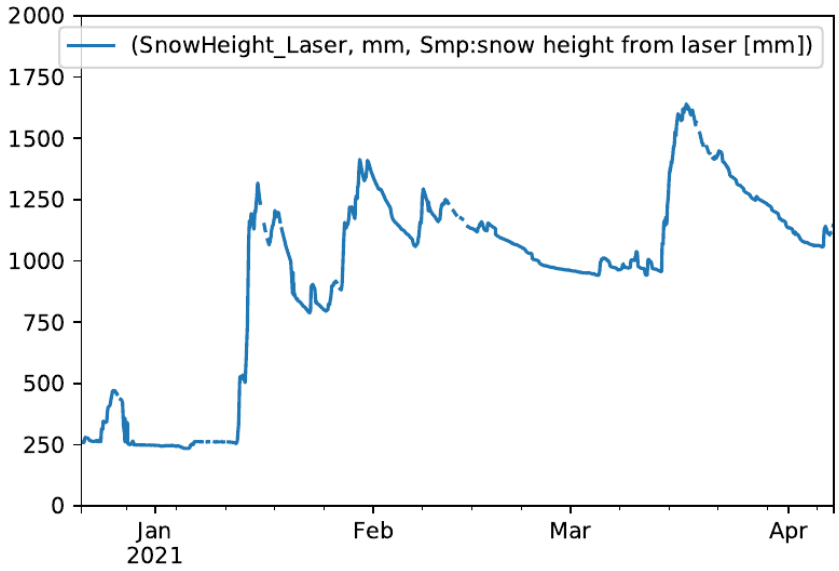
Wind speed DAV



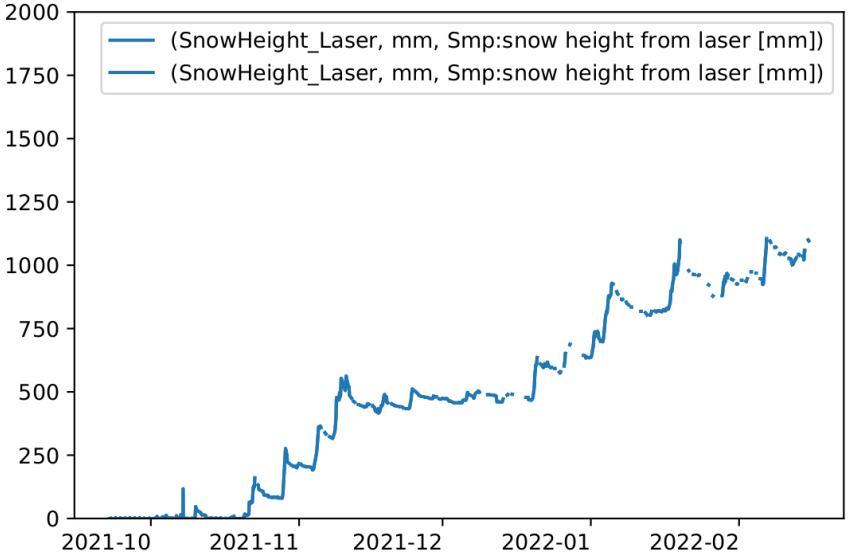
Wind speed TAJ



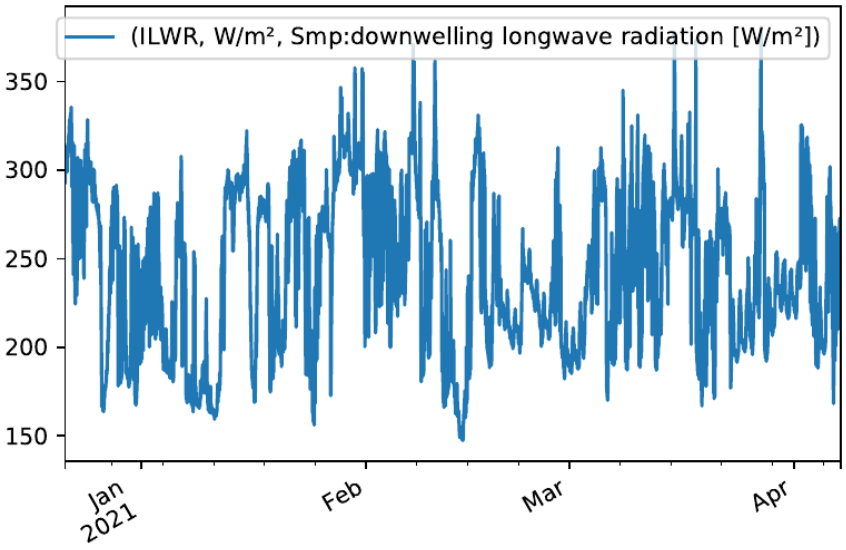
Snow height DAV



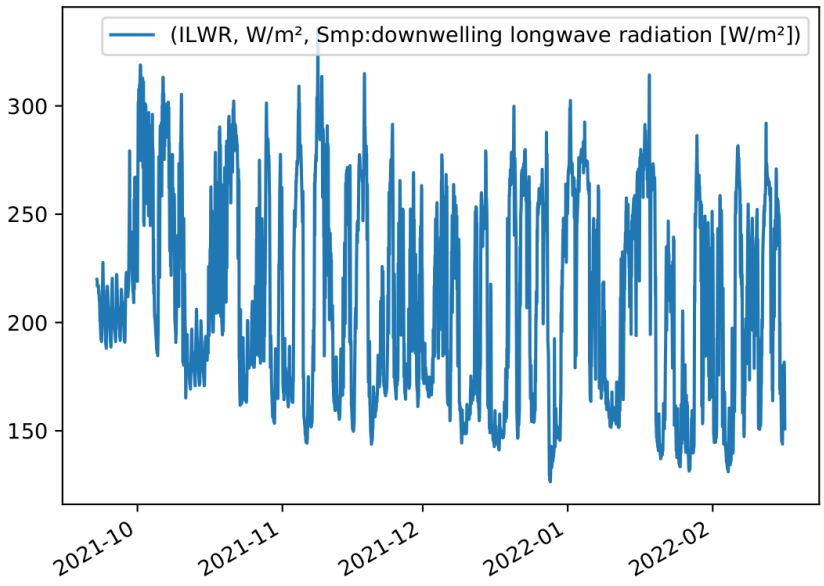
SnowHeight TAJ



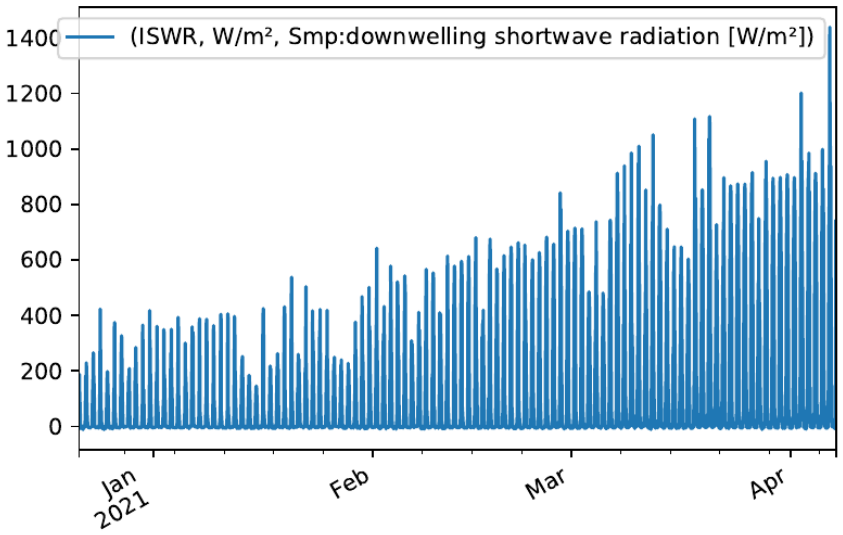
ILWR DAV



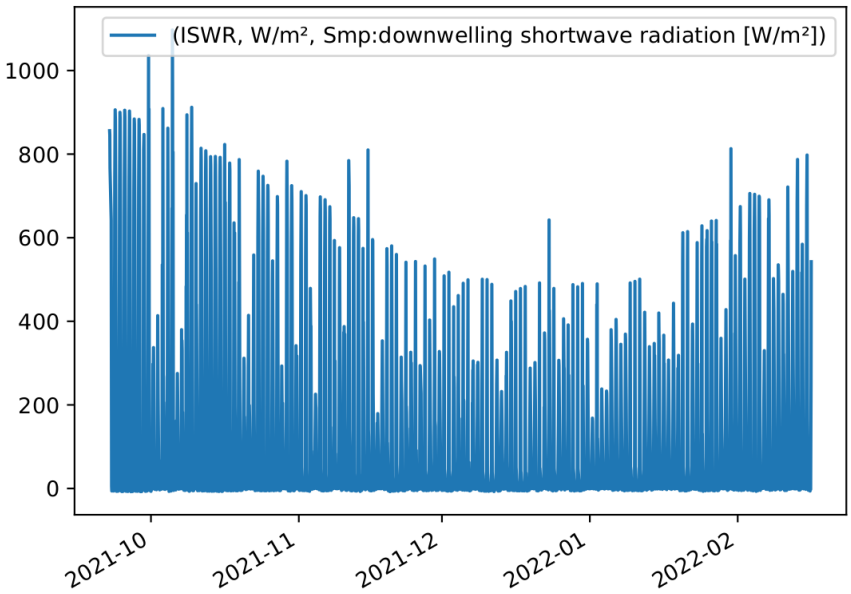
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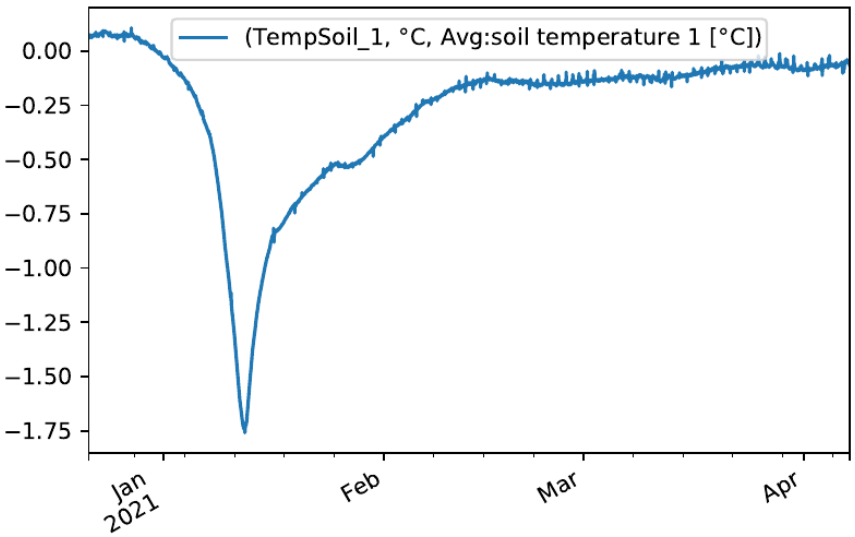
ISWR height DAV



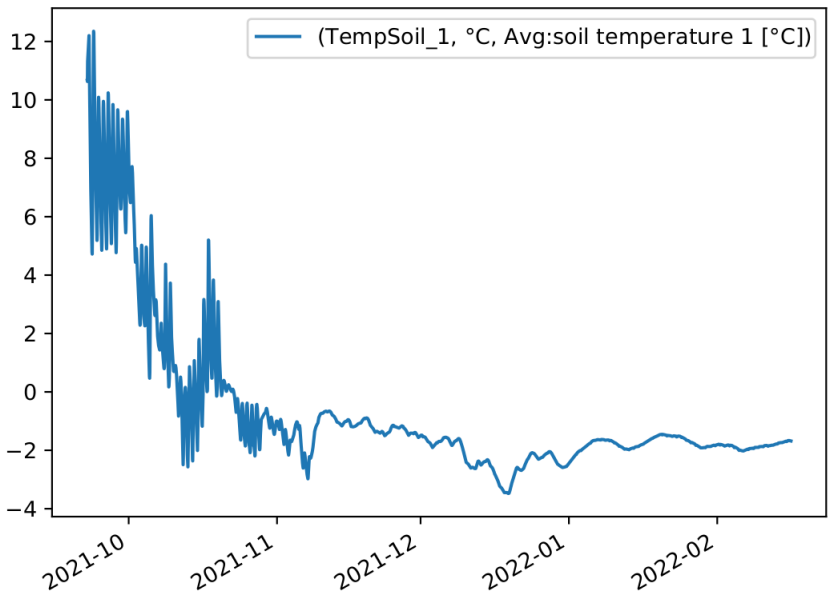
ISWR TAJ



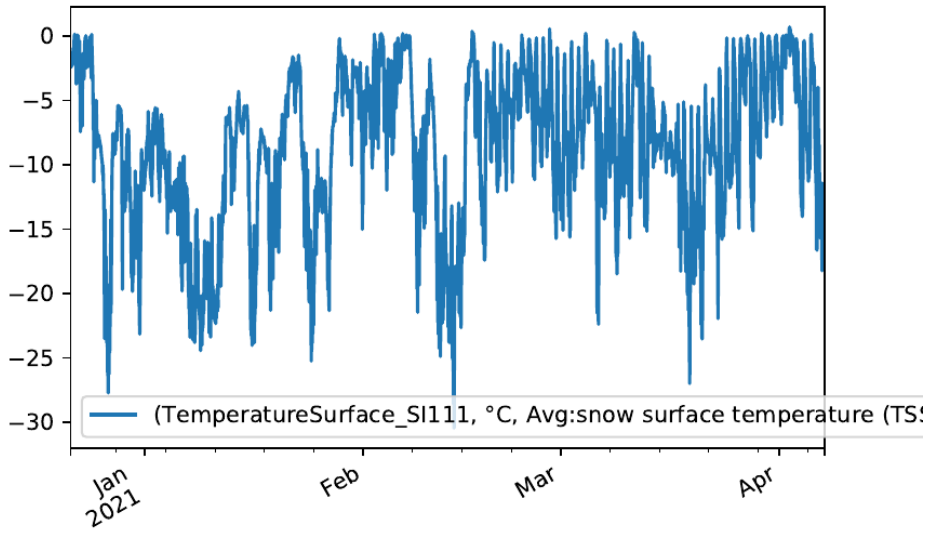
Soil Temperature 10cm DAV



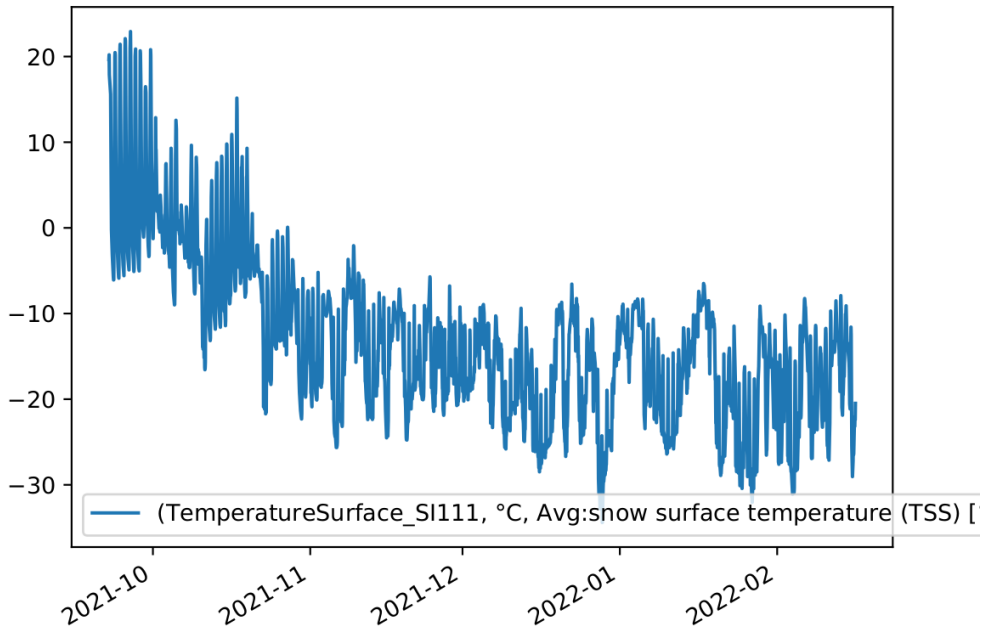
Soil Temperature 10cm TAJ



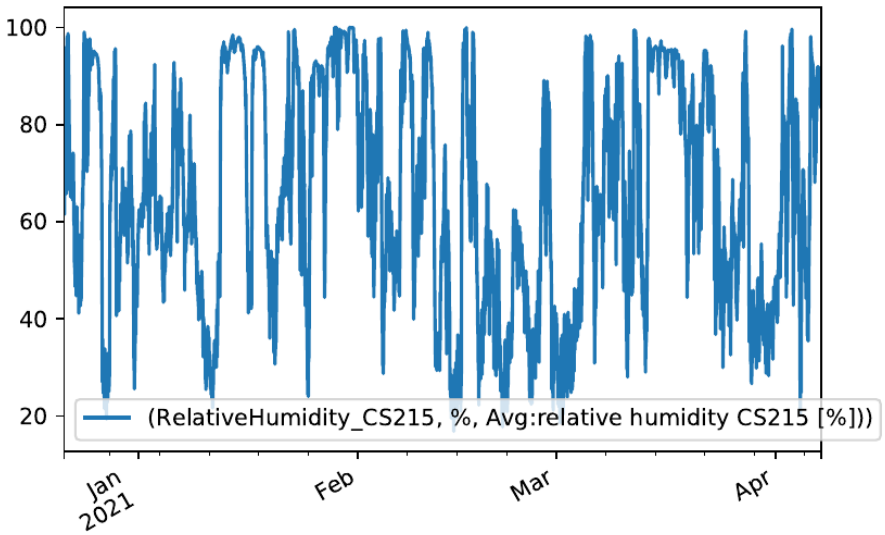
Surface temperature DAV



Surface temperature TAJ



Relative humidity DAV



Relative humidity TAJ

